METRIC
DOD-N-63548
17 March 1982

# MILITARY SPECIFICATION

## NUT, PLAIN, HEXAGON, WHEEL MOUNTING, METRIC GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers metrically dimensioned steel hexagon plain wheel mounting nuts.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)
QQ-Z-325 - Zinc Coating, (Electrodeposited), Requirements for
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

MILITARY

MIL-I-6868 - Inspection Process, Magnetic Particle
DOD-N-63548/1 - Nut, Plain, Hexagon, Cone Seat, Wheel Mounting, Metric

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5310

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STANDARDS

FEDERAL

FED-STD-H28/21 - Screw-Thread Standards for Federal Services, Section 21, Metric Screw Threads
FED-STD-66 - Steel: Chemical Composition and Hardenability

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-1312 - Fasteners, Test Methods

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 563M - Standard Specification for Carbon and Alloy Steel Nuts

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) HANDBOOK

SAE J122a - Surface Discontinuities of Nuts

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references other than specification sheets cited herein, the text of this specification shall take precedence.



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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Recycled and reclaimed materials shall be used to the maximum extent practicable.

3.2.1 Carbon steel. The nuts shall be made of carbon steel in accordance with FED-STD-66 composition 1151 or ASTM A 563M property class 9.

3.3 Hardness. Nut hardness shall not exceed 30HRC.

3.4 Protective finish. Protective finishes shall be as follows:

3.4.1 Cadmium plating. Nuts shall be cadmium plated in accordance with QQ-P-416, type II, class 2 (7.6 µm thick).

3.4.2 Zinc coating. Nuts shall be zinc coated in accordance with QQ-Z-325, type II, class 2 (13 µm thick).

3.4.3 Hydrogen embrittlement. Cadmium plated carbon steel nuts shall be subjected to an embrittlement relief treatment within four hours after plating in accordance with QQ-P-416.

3.5 Design.

3.5.1 Threads. Nut threads shall conform to the "M" profile, tolerance class 6H, as defined in FED-STD-H28/21.

3.5.2 Concentricity. The thread axis shall be concentric with axis of the nut body within a tolerance equal to 3 percent for all hexagon nuts.

3.6 Dimensions. Dimensions and tolerances shall be in accordance with the applicable specification sheet (see 3.1) and as specified herein.



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3.7 Discontinuities. The depth of discontinuities such as laps, seams and inclusions in the nuts shall be no greater than 3 percent of the nominal thread size. The length of discontinuities shall not exceed the height of the nut.

3.7.1 Cracks. Quenched cracks are not permitted in any location. Forging cracks located in the top and bottom face are permitted provided: no more than two forging cracks extend from the thread across the full width of the face; no forging crack extends into the thread beyond the first full thread; and the width of any forging crack does not exceed 0.20 mm plus 0.01D, where D is the nominal nut thread size in mm. Forging cracks located at the intersection of the top or bottom face with a flat are permitted provided that the width does not exceed 0.25 mm plus 0.020D, where D is the nominal nut thread size in mm. The depth of forging cracks shall be not greater than 3 percent of the nominal nut thread size in mm.

3.8 Proof load. Nuts shall withstand the proof loads specified in table I without rupture of metal or thread stripping. Upon removal of the proof load, nuts shall be manually removeable from a mating bolt or mandrel without assistance of a tool except for the initial one-half turn of the nut.

TABLE I. Proof loads.

| Thread size | M12 | M20 | M22 |
|---|---|---|---|
| Proof load, kN (min) | 80 | 225 | 279 |
| Stress area ($mm^2$) (nom) | 84.3 | 245 | 303 |
| Proof load stress (MPa) (min) | 950 | 920 | 920 |

3.9 Marking. Marking of the nut shall be in accordance with ASTM A 563-M and MIL-STD-130 as applicable.

3.10 Workmanship. Finished nuts shall be uniform in quality and appearance. Surfaces shall be free of foreign matter, burrs, loose scale and any other defects which may adversely affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Material inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed at ambient temperature, humidity and atmospheric pressure.

4.4 Material inspection. Material inspection shall consist of certification supported by verifying data that materials used in fabricating nuts are in accordance with 3.2 prior to such fabrication.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table II.

TABLE II. Quality conformance inspection.

| Inspection | Requirement paragraph | Test method paragraph |
|---|---|---|
| Group A | | |
| Design | 3.5 | 4.6.1 |
| Dimensions | 3.6 | 4.6.1 |
| Workmanship | 3.10 | 4.6.1 |
| Group B | | |
| Discontinuities | 3.7 | 4.6.1,4.6.3 |
| Hardness | 3.3 | 4.6.2 |
| Proof load | 3.8 | 4.6.4 |

4.5.1 Inspection lot. An inspection lot shall consist of completed nuts covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.5.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.3 Sampling for group A inspection. Sampling shall be in accordance with inspection level S-3 of MIL-STD-105 with an acceptable quality level (AQL) of 2.5 percent defective for major defects. For minor defects, inspection level S-3 and an AQL of 4.0 percent defective shall apply.



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4.5.4 Sampling for group B inspection. Samples for tests shall be selected from lots which have been visually and dimensionally examined and accepted. Sampling shall be in accordance with MIL-STD-105 with inspection level and AQL as follows:

| Test | Inspection level | AQL |
|---|---|---|
| Discontinuities (magnetic particle) | S-4 | 1.5 |
| Proof load | S-4 | 1.5 |
| Hardness | S-2 | 4.0 |

4.5.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of PPP-H-1581.

4.6 Methods of inspection.

4.6.1 Examination. Samples taken as specified in 4.5.3 and 4.5.4 shall be thoroughly examined to determine conformance with this specification and applicable specification sheet. Examination shall be conducted in accordance with table III.

TABLE III. Classification of defects.

| Category | Defect | Inspection method |
|---|---|---|
| Critical | None defined | |
| Major | | |
| 101 | Threads not as specified (see 3.5.1) | SIE 1/ |
| 103 | Evidence of cracks, laps, seams, inclusions (see 3.7) | Visual and SIE |
| 104 | Pilot dimensions and concentricity not as specified (see 3.5.2, 3.6) | SIE |
| Minor | | |
| 201 | Nut thickness not as specified (see 3.6) | SIE |
| 202 | Other dimensions not as specified (see 3.6) | SIE |
| 203 | Workmanship not as specified (see 3.10) | Visual |

1/ Standard inspection equipment.

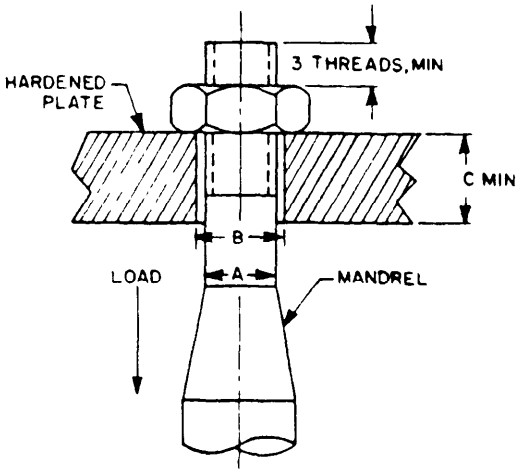
4.6.2 Hardness. When specified (see 6.2), sample nuts taken as specified in 4.5.4 shall be tested for hardness in accordance with MIL-STD-1312, test method 6.



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4.6.3 Discontinuities. When specified (see 6.2), sample nuts taken as specified in 4.5.4 shall be subjected to magnetic particle inspection in accordance with MIL-I-6868 for conformance to 3.7. When specified (see 6.2), samples with seams or potentially deep forging cracks shall be subjected to the cone proof load test of SAE J122a.

4.6.4 Proof load. Samples taken as specified in 4.5.4 shall be mounted on a mandrel in accordance with figure 1, and loaded as specified in 3.8, table I. Mandrel hardness shall be 45HRC, minimum, and threads shall be class 5h6g in accordance with FED-STD-H28/21, except that the maximum major diameter shall be the minimum major diameter plus 0.25 times the class 6g tolerance range. The nut shall withstand the specified load, applied axially and in tension, without rupture or stripping.




| DIMENSIONS | | |
|---|---|---|
| A<br>Thread size<br>(mm) | B<br><br>(mm) | C<br>min<br>(mm) |
| M12<br>M20<br>M22 | 12.050+.110<br>20.065+.130<br>22.065+.130 | 12<br>20<br>22 |

FIGURE 1. Fixture for proof load test.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2)

6. NOTES

6.1 Intended use. Wheel mounting nuts are intended for mounting wheel rims on automotive vehicles such as automobiles and trucks.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.
b. Applicable specification sheet part number (see 3.1)
c. Tests if required (see 4.6.2, 4.6.3)
d. Level (degree) of protection in accordance with PPP-H-1581, ordering date (see 5.1)



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6.3 Definitions.

6.3.1 Crack. A crack is a clear crystalline fracture passing through or across the grain boundaries without inclusion of foreign elements.

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Army - AR
Air Force - 99

Review activities:
Army - AT, EA
DLA - IS
NSA - NS

User activities:
Army - ME
Navy - MC

Agent:
DLA - IS

Preparing activity:
Army - AR

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